

Question Number	Answer	Mark
2(a)	The only correct answer is B capsule and pili <i>A is not correct because flagellum are not involved in adhesion</i> <i>C is not correct because flagellum are not involved in adhesion</i> <i>D is not correct because flagellum are not involved in adhesion</i>	(1)

Question Number	Answer	Additional guidance	Mark
2(b)(i)	An explanation that makes reference to two of the following points: • will not run out / more plants can be grown / renewable / available to future generations (1) • biodegradable / can be broken down by {decomposers / bacteria / fungi} (more rapidly) (1) • carbon neutral (1)	ignore decompose unqualified	(2)

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	An answer that makes reference to the following points: - there is a reduction in bacteria with poplar, pine and polystyrene / there is a reduction in number of bacteria with all packaging (1) - decrease in bacteria is due to antimicrobial properties (1) - pine {is the most effective (packaging) / has most antimicrobial properties} (1) - relevant comment on {SD / sample size / suitable variable} (1)	accept correct statement about decrease in numbers of bacteria for one packaging type e.g. pine has greatest decrease / plant-based has bigger decrease / oil-based have insignificant decrease accept {poplar / pine / polystyrene} (packaging) {have antimicrobial properties / produce chemicals which kill bacteria} / plant-based products have antimicrobial properties accept a given antimicrobial property e.g. low pH accept {plant-based / wood} packaging is more effective (than oil-based) accept pine (packaging) is more effective than poplar (at reducing bacterial growth) ignore HDPE is least effective e.g. {significant difference between the / valid} results due to no overlapping of SD / poplar wood results have lower reliability due to largest SD ignore {lower reliability / validity} due to only 4 groups accept no information on how many fish were in each group / no information on {type/age} of fish	(4)